Results

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0%

0
Out of 0 points

26:56
Time for this attempt

Your Answers:

1 0 / 0 points

A neural network that has 5 neurons in the hidden layer and 2 neurons in the output layer, and takes a 10-dimensional input vector. How many trainable parameters (weights and biases) does this network have? Enter the answer as a single 2 digit number.

Answer: 67 trainable parameters

2 0 / 0 points

The length of sequence of words is 20, the embedding size of each word is 100.

Self attention weights are calculated using the following formula:

$soft\max\left(\frac{Q\times K^T}{\sqrt{d_k}}\right)\times V$ where Q, K, V is query, key, value, all have (#words x dimension of embedding).

What is the size of the resultant SelfAttention(Q;K;V)?

- 100 x 100
- 3 scalar values
- 100 x 20
- 20 x 100

3 0 / 0 points

The IR system retrieved 5 documents, second and fourth were relevant, others were not relevant. This can be represented as 01010, where from left to right in increasing rank, a "1" represents relevant document. Calculate the MAP score.

- 0.375
- 0.12
- 0.5
- 0.2

4 0 / 0 points

Evaluate the following expression to True or False

$\neg(T\wedge F)\vee T\Rightarrow T$

- False
- True
- The expression can not be evaluated

5 0 / 0 points

What is "masking" in the context of transformer-based NLP models? Select one most correct answer.

- Replacing certain words in the input with random words to prevent the model from overfitting.
- Replacing certain words in the input with a special token to prevent the model from seeing them.
- Replacing certain words in the input and output with a special token to prevent the model from seeing or predicting them.
- Replacing certain words in the output with a special token to prevent the model from predicting them.

6 0 / 0 points

Given the following text: "time after time after time and again", calculate PMI for the "after time" bi-gram. Enter the value in the field provided. Round to two digits after the decimal point, e.g. 9.91

1.44

7 0 / 0 points

From example in section 17.4.6 (Jurafsky textbook) calculate the probability of "Janet/NNP will/MD". Use tables 17.12, 17.13 and equation 17.17 for probability calculation.

	NNP	MD	VB	JJ	NN	RB	DT
<s>	0.2767	0.0006	0.0031	0.0453	0.0449	0.0510	0.2026
NNP	0.3777	0.0110	0.0009	0.0084	0.0584	0.0090	0.0025
MD	0.0008	0.0002	0.7968	0.0005	0.0008	0.1698	0.0041
VB	0.0322	0.0005	0.0050	0.0837	0.0615	0.0514	0.2231
JJ	0.0366	0.0004	0.0001	0.0733	0.4509	0.0036	0.0036
NN	0.0096	0.0176	0.0014	0.0086	0.1216	0.0177	0.0068
RB	0.0068	0.0102	0.1011	0.1012	0.0120	0.0728	0.0479
DT	0.1147	0.0021	0.0002	0.2157	0.4744	0.0102	0.0017

Figure 17.12 The A transition probabilities $P(t_i|t_{i-1})$ computed from the WSJ corpus without smoothing. Rows are labeled with the conditioning event; thus $P(VB|MD)$ is 0.7968. <s> is the start token.

	Janet	will	back	the	bill
NNP	0.000032	0	0	0.000048	0
MD	0	0.308431	0	0	0
VB	0	0.000028	0.000672	0	0.000028
JJ	0	0	0.000340	0	0
NN	0	0.000200	0.000223	0	0.002337
RB	0	0	0.010446	0	0
DT	0	0	0	0.506099	0

Figure 17.13 Observation likelihoods B computed from the WSJ corpus without smoothing, simplified slightly.

$$\hat{t}_{1:n} = \underset{t_1\ldots t_n}{\operatorname{argmax}} P(t_1\ldots t_n|w_1\ldots w_n) \approx \underset{t_1\ldots t_n}{\operatorname{argmax}} \prod_{i=1}^{n} \overbrace{P(w_i|t_i)}^{\text{emission}} \overbrace{P(t_i|t_{i-1})}^{\text{transition}} \qquad (17.17)$$

Select the closest result to your probability (because precision may vary)

- 7.9e-9
- 9.9e-6
- 0.11
- 3.0e-8

8 0 / 0 points

A two-class model was trained and then tested with a dataset of 200 instances. The test set contained 150 instances in negative class N, and 50 instances in positive class P. As a result of testing, the following counts were obtained:

100 instances of N were classified correctly,
30 instances of N were classified into P,
20 instances of P were classified correctly,
50 instances of P were classified into N.

Calculate precision p, recall r and F1 score for the positive P class. Then choose the correct answer from below.

- ( ) p=0.17, r=0.77, F1=0.27
- ( ) p=0.6, r=0.29, F1=0.39
- ( ) p=0.83 r=0.25, F1=0.38
- (●) p=0.4, r=0.29, F1=0.33

9 0 / 0 points

What is the dialog policy in GUS system? **Select all correct answer.**

- [x] Keep asking questions until the frame is full
- [x] Decide when to ask a clarification question
- [ ] Decide what to ask next
- [x] Decide when to finish conversation

10 0 / 0 points

The longest common sequence (LCS) can be used to calculate the edit distance between two texts. This calculation counts how many addition and deletion operation need to be performed to make text1 the same as text2.

Given two words w1="kitten" and w2="sitting", calculate LCS edit distance. Enter the distance in the space provided.

5

11 0 / 0 points

What is the main purpose of the activation function in a neural network?

- ( ) To extend the architecture of the network to multiple layers
- (●) To introduce non-linearity to the model
- ( ) To calculate the gradients during training
- ( ) To prevent the network from overfitting

12 0 / 0 points

Calculate the probability of sequence “I want Chinese food”, based on probabilities of bigrams given below.

Tip: use the example below Fig 3.2 in Jurafsky's textbook.

P("i" |<s>) = 0.25, P(</s>| "food") = 0.68

	i	want	to	eat	chinese	food	lunch	spend
i	0.002	0.33	0	0.0036	0	0	0	0.00079
want	0.0022	0	0.66	0.0011	0.0065	0.0065	0.0054	0.0011
to	0.00083	0	0.0017	0.28	0.00083	0	0.0025	0.087
eat	0	0	0.0027	0	0.021	0.0027	0.056	0
chinese	0.0063	0	0	0	0	0.52	0.0063	0
food	0.014	0	0.014	0	0.00092	0.0037	0	0
lunch	0.0059	0	0	0	0	0.0029	0	0
spend	0.0036	0	0.0036	0	0	0	0	0

Figure 3.2 Bigram probabilities for eight words in the Berkeley Restaurant Project corpus of 9332 sentences. Zero probabilities are in gray.

The above table reads as follow, e.g. P("to"|"want")=0.66

Download accessible format.

- ( ) 0.0
- ( ) 0.00019
- (●) 0.000279
- ( ) 3.35478E-07